VELOOP REWARDS - Referral & Engagement Analytics

Data Analyst Internship, Task 01 | Work Package E

Authenticity notice: Every field, event and measure below is **Proposed** or **To Be Verified**. No targets or thresholds are invented, and no simulated or real VELOOP statistics appear here.

1. Purpose and key principles

- **Registration is not conversion.** The referral funnel has four separate stages, each with its own event and timestamp.
- **One activation definition.** Referral activation should use the same definition as user activation in the lifecycle model, otherwise two parts of the product will report different numbers (A010).
- **Active means qualifying activity.** Logging in or viewing a screen is not enough. Section 5 lists which events are recommended and which need a decision.
- **Definitions can change.** activation_rule_version records which rule applied, so older results remain explainable.

2. Referral funnel (E0)

Stage	Meaning	Event	Timestamp Field	Measure (formula only, no targets)	Caution	Status
1. Referral Created	A referral link or code is generated or shared by an existing user	referral_created	created_at	Referrals created = COUNT(referral_id) by created_at	Creating a code once is not the same as sharing it; define what counts as created	Proposed
2. User Registered	A new user signs up using that referral	referral_registered	registered_at	Registration rate = referrals registered / referrals created	Registration is NOT a successful referral	Proposed
3. User Activated	The referred user does the agreed activating action	referral_activated	activated_at	Activation rate = referrals activated / referrals registered	Needs one agreed activation definition shared with the user lifecycle (A010)	To Be Verified
4. Referral Converted	The referral meets the programme's success condition	referral_converted	converted_at	Conversion rate = referrals converted / referrals created, for referrals created in the period	Recent referrals look low until they mature; show alongside open referrals	Proposed

3. Referral data (E1)

Field	Type	Required?	Definition	Validation	Status
referral_id	string	Yes	Unique key for one referral relationship	Unique; immutable	Proposed
referring_user_id	string	Yes	User who invited	Must exist in Users	To Be Verified
referred_user_id	string	Conditional	User who signed up through the referral; empty until registration	Must exist in Users; must differ from referring_user_id	To Be Verified
referral_code	string	Optional	Code or link token used	Unique per referring user	To Be Verified
created_at	timestamp (UTC)	Yes	When the referral was created	ISO 8601 UTC; not in the future	Proposed
registered_at	timestamp (UTC)	Conditional	When the referred user registered	Not before created_at; equals referred user's registered_at	Proposed
activated_at	timestamp (UTC)	Conditional	When the referred user met the activation rule	Not before registered_at	Proposed
converted_at	timestamp (UTC)	Conditional	When the referral met the success condition	Not before activated_at, unless the rule says otherwise (to be verified)	Proposed
status	string (enum)	Yes	Current stage of the referral	created, registered, activated, converted, closed (final list to be agreed)	Proposed
source_channel	string (enum)	Optional	How the referral was shared (link, code, other)	Approved list	Proposed
activation_rule_version	string	Optional	Version of the activation definition applied, so past results stay explainable when the rule changes	Matches the change log	Proposed
referral_reward_id	string	Optional	Reward given for this referral, if any	Must exist in Rewards	To Be Verified
closed_reason_category	string (enum)	Conditional	Why a referral was closed without converting, at category level only	No free text; categories to be agreed; no fraud rules invented	To Be Verified

Referral stage transitions

Disallowed transitions become data quality checks in the later quality framework.

From	To	Allowed?	Note
created	registered	Yes	Normal path
registered	activated	Yes	Normal path
activated	converted	Yes	Normal path
registered	converted	Only if the programme has no separate activation step	To be verified
created	activated	No	Cannot activate before registering
converted	registered	No	Cannot go backwards
registered	registered	No	Duplicate registration for one referral; data quality error
any	referred_user_id = referring_user_id	No	Self-referral is a data integrity error

Referral measures

Full KPI definitions are completed in the KPI Catalogue (Work Package I). These are the measures it will build on.

Measure	Definition	Formula	Caution	Status
Referrals Created	Referrals created in the period	COUNT(referral_id) by created_at	Counts relationships, not unique referrers	Proposed
Referral Registrations	Referred users who registered	COUNT(referral_id) where registered_at in period	Registration is not conversion	Proposed
Referral Activations	Referred users who activated	COUNT(referral_id) where activated_at in period	Depends on the agreed activation rule	To Be Verified
Referral Conversions	Referrals that converted	COUNT(referral_id) where converted_at in period	Period by conversion date differs from cohort view; label which is used	Proposed
Registration Rate	Share of created referrals that registered	registered / created for referrals created in the period	Matures over time	Proposed
Activation Rate	Share of registered referrals that activated	activated / registered for referrals registered in the period	Matures over time	To Be Verified
Conversion Rate	Share of created referrals that converted	converted / created for referrals created in the period	Show open referrals beside it	Proposed
Time to Activate	How long from registration to activation	Median(activated_at - registered_at)	Excludes referrals not yet activated	Proposed
Active Referrers	Users who created at least one referral in the period	COUNT(DISTINCT referring_user_id) by created_at	Does not show quality of referrals	Proposed

4. Engagement measurement (E2)

Which events count as activity

Daily, weekly and monthly active users all depend on one list of qualifying events. This is a stakeholder decision; the recommendation below is a starting point.

Event	Include in 'active'?	Reason	Status
task_completed	Recommended: yes	Validated earning activity	Proposed
daily_reward_claimed	Recommended: yes	Deliberate daily action	Proposed
spin_completed	Recommended: yes	Feature use that finished	Proposed
mission_completed	Recommended: yes	Feature use that finished	Proposed
mining_claimed	Recommended: yes	Return behaviour	Proposed
withdrawal_requested	Recommended: yes	Deliberate wallet action	Proposed
task_started	Decision needed	Intent, but not an outcome	To Be Verified
user_logged_in	Decision needed	Opening the app only; recommended to track separately as 'visits'	To Be Verified
task_viewed / feature_opened	Recommended: no	Passive; can be automatic	Proposed
reward_credited	Recommended: no	System action, not a user action	Proposed

Engagement measures

Measure	Definition	Formula	Events / Fields Needed	Dimensions	Caution	Status
Daily Active Users (DAU)	Unique users with at least one qualifying event on a calendar day	COUNT(DISTINCT user_id) where qualifying event on date	qualifying events, user_id, event timestamp	platform, registration_source, lifecycle_stage	Calendar day depends on the agreed timezone (A012)	Proposed
Weekly / Monthly Active Users	Same, over a week or month	COUNT(DISTINCT user_id) in the window	as above	as above	Window boundaries must be fixed and documented	Proposed
Returning Users	Active users in the period whose first qualifying activity was before the period	COUNT(DISTINCT user_id) active in period AND first_activity_at < period start	user_id, first_activity_at	platform, cohort month	Needs first_activity_at from the user model	Proposed
New Active Users	Users whose first qualifying activity falls in the period	COUNT(user_id) where first_activity_at in period	first_activity_at	registration_source	Registered is not the same as active	Proposed
Feature Adoption	Share of active users who used a feature at least once	users with the feature's start event / active users in period	feature start events (e.g. spin_started, mission_started)	feature, platform	Choose one denominator and keep it stable	Proposed
Feature Frequency	How often feature users use the feature	feature events / users of the feature in the period	feature events, user_id	feature, lifecycle_stage	Skewed by heavy users; show median and mean	Proposed
Repeat Usage	Users who return to a feature on separate days	users with the feature on 2 or more distinct days / users with 1 or more days	feature events, event date	feature, cohort	Window length to be agreed	Proposed
Task Activity	Task starts, validated completions and failures per active user	COUNT(task_started), COUNT(task_completed), COUNT(task_failed) / active users	task_started, task_completed, task_failed	task_type, feature	Completion needs validation (see Work Package C)	Proposed
Daily Activity Depth	Qualifying events per active user per day	qualifying events / DAU	qualifying events	platform, feature	Depends on which events qualify	Proposed
Cohort Retention	Share of a registration cohort active again after N days	users from cohort active on day N / cohort size	registered_at, qualifying events	registration_source, platform	Needs stable user_id (A003); N values to be agreed	Proposed
Stickiness	Relationship between daily and monthly activity	DAU / MAU for the same period	DAU, MAU	platform	A ratio; no target is set here	Proposed

5. Open questions for verification

- What exactly makes a referred user 'activated', and does it match the user lifecycle definition? (A010)
- Can a user have only one referrer? Can a referral be closed or disqualified, and by what category?
- Is the referral reward given at registration, activation or conversion? (A002, A013)
- Which events qualify a user as active, and which timezone defines a day? (stakeholder decision, A012)
- Do features outside earning (Help Center, wallet view) need a generic feature_opened event (name to be agreed) to measure adoption?
- How long is the activity and retention window? (no value invented here)

Next: Work Package F - Production Event Tracking Plan, one of the most important deliverables.